COMSATS UNIVERSITY ISLAMABAD
Lahore Campus
Department of Computer Science

■ Lab Midterm Exam

Course Title:	DevOps for Cloud Computing			Course Code:	CSC418	Credit	3(2,1)
Course Instructor/s:	Dr. M. Hasanain Ch			Program Name:	BS(CS)		
Semester:	6th & 7th	Batch:	FA23	Date:	06 April 2026		
Time Allowed:	1 Hour 30 Minutes			**Maximum Marks:**		**25**	

Q1. [CLO: 5; Bloom Taxonomy Level <Applying> Apply DevOps pipeline automation techniques for code deployment.]

[Marks: 10 + 10 + 5 = 25]

You are required to design and implement a complete Cloud Computing Deployment Pipeline that takes a working application from local development all the way to cloud hosting using modern DevOps tools and platforms

Your pipeline must demonstrate:

1. Application Development & Local Testing
2. Containerization using Docker
3. Image Publishing to Docker Hub
4. Cloud Deployment using Azure Kubernetes Service (AKS)
5. Version Control using GitHub Commands

Project Requirement:

Each student must select a semester project that demonstrates practical DevOps implementation and includes the following components:

1. **Front-End Interface:**
   A user-facing interface (e.g., built using HTML/CSS/JS, React, Angular, etc.) to interact with the application.

2. **Back-End Functionality:**
   A functioning server-side application (e.g., Node.js, Python Flask/Django, Java Spring Boot, etc.) that processes requests and connects with the database.

3. **Database Integration:**
   A database (e.g., MySQL, MongoDB, PostgreSQL, etc.) must be used for storing and retrieving data dynamically.

1: Dockerization & Local Deployment (10 Marks)

You are required to take any semester project of your choice (Node.js or Python recommended) and perform the following tasks:

- Set up your application and ensure it runs locally without errors. (2 Marks)
- Create a Dockerfile based on your application. (3 Marks)
- Build a custom Docker image successfully and run the application inside a Docker container. (3 Marks)
- Push the image to Docker hub repository using commands. (2 Marks)

2: Azure Kubernetes Cloud Deployment (10 Marks)

Deploy your Docker image to **Azure Kubernetes Service** using Portal-based deployment:

- Create an Azure Kubernetes Cluster (AKS). (3 Marks)
- Deploy containerized application from Docker Hub onto AKS. (4 Marks)
- Expose the app using a public IP address and provide a reachable link. (3 Marks)

3: GitHub Repository & Commands Usage (5 Marks)

Use GitHub commands for this task as learned in the lab

- Create a GitHub repository (public or private). (1 Mark)
- Add all your project files including Dockerfile. (2 Marks)
- Use Git commands: add, commit, push, pull properly. (2 Marks)

Required Submission:

Submit a word/pdf file that includes:

- GitHub Repository Link

- Docker Hub Image Link
- Azure App Public URL
- Screenshots of Docker and AKS deployments